



CSE-476: Introduction to Natural Language Processing  
Quiz 1

## Multiple Choice Questions (Single correct)

*Instructions: Select the correct option with a short explanation.*

- Q1. In the BIO tagging scheme for NER, what does the “B-” prefix indicate?
- (A) Beginning of an entity
  - (B) Inside an entity
  - (C) Outside any entity
  - (D) Between two tokens
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- (A) It uses only words (observations) and ignores labels
  - (B) It is identical to an HMM
  - (C) It models transitions between labels but does not consider the words
  - (D) It models emissions  $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- (A)  $P(\text{label} \mid \text{previous label})$
  - (B)  $P(\text{label} \mid \text{word})$
  - (C)  $P(\text{word} \mid \text{previous word})$
  - (D)  $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes Semantic Role Labeling (SRL)?:
- (A) A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
  - (B) A process that assigns grammatical roles like subject and object based on syntactic structure
  - (C) A method that classifies named entities and links them to a knowledge base
  - (D) A technique that labels words with their parts of speech and morphological features

- Q5. Which of the following best describes Byte Pair Encoding (BPE)?
- (A) A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
  - (B) A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
  - (C) A tokenization method that splits text based on whitespace and punctuation
  - (D) A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of tokenization in NLP?
- (A) To label each token with a grammatical category
  - (B) To convert text into a sequence of symbols suitable for modeling
  - (C) To reduce vocabulary size to zero
  - (D) To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer?
- (A) It produces too many tokens
  - (B) It does not work well for languages without explicit word boundaries
  - (C) It cannot tokenize English at all
  - (D) It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task?
- (A) Part-of-speech tagging
  - (B) Named Entity Recognition
  - (C) Dependency parsing
  - (D) Sentiment analysis
- Q9. What is a common limitation of BIO tagging for Named Entity Recognition?
- (A) It cannot represent overlapping or nested entities
  - (B) It requires too many labels
  - (C) It does not support multiple entity types
  - (D) It ignores entity boundaries

Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?

- (A) Speed vs accuracy
- (B) Tokenized sequence length vs vocabulary size
- (C) Precision vs recall
- (D) Memory vs computation

*End of Paper*